Tairan Chen

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EDUCATION

Stevens Institute of Technology — ***Hoboken, NJ, USA***

Ph.D. in Computer Science — *Sept. 2025 - Present*

Advisor: Prof. Hui Wang

Research Focus: Deep Learning, Large Language Models, Trustworthy Machine Learning

Jilin University — ***Jilin, China***

Bachelor in Computer Science and Technology — *Sept. 2020 - Jun. 2024*

Overall GPA : 86.01 / 100

Related Coursework: C Programming (95), Compiler Principles (94), Computer Organization (90), Digital Logic Circuits (90), Microcomputer Systems (89), Discrete Mathematics (89), Data Structures (87), Operating System(86)

WORK EXPERIENCE

Research Engineer — **FlashIntel**

FlashLabs — *Mar. 2025 - Sept. 2025*

Shanghai, China · On-site

Focus: Large Language Models, Text-to-Speech Synthesis, End-to-End Spoken Dialogue Systems

Research Assistant — **Tsinghua University**

Tsinghua SAIL Group — *Dec. 2023 - Feb. 2025*

Advisors: Prof. Jun Zhu and Associate Prof. Hang Su, Department of Computer Science and Technology

Research Focus: Computer Vision, Natural Language Processing, Adversarial Machine Learning

RESEARCH EXPERIENCE

Chroma 1.0: Real-Time End-to-End Spoken Dialogue Model with Voice Cloning — **FlashIntel**

Researcher (Equal Contribution) — *Mar. 2025 - Sept. 2025*

FlashLabs, Shanghai, China

Background: End-to-end spoken dialogue models struggle with combining low-latency streaming and high-fidelity personalized voice cloning. Existing systems either sacrifice speaker identity for real-time performance or lack streaming capability. We proposed Chroma 1.0, the first open-source, real-time end-to-end spoken dialogue model achieving both goals simultaneously.

- Designed and implemented a streaming multimodal LLM architecture integrating a reasoning model (**Qwen2.5-Omni**) and a **LLaMA-based backbone (1B)**, enabling synchronized text-audio generation via interleaved token scheduling (text:audio = 1:2).
- Built core model components including a **100M-parameter audio decoder** for frame-synchronous residual codebook prediction and a **neural codec decoder** (causal CNN) for high-fidelity waveform reconstruction.
- Achieved real-time inference with **RTF = 0.43** (2.3× faster than real-time) and reduced latency to **TTFT = 146.87ms** through system-level optimization and streaming design.
- Improved voice cloning quality with **SIM = 0.817** (**+10.96%** over human baseline), outperforming Seed-TTS, F5-TTS, CosyVoice 3, and FireRedTTS-2 on CommonVoice.
- Demonstrated strong reasoning and dialogue capabilities with a **4B-parameter** model, ranking **top-2** on URO-Bench against models up to 9B.

Achievement: Published on **arXiv**: arXiv:2601.11141. Code and models publicly released at GitHub and HuggingFace.

Active Perception Defense Against 3D Adversarial Patches — **Tsinghua University**

Research Assistant — *Apr. 2024 - Present*

Advisor: Associate Prof. **Hang Su** in the Department of Computer Science and Technology at Tsinghua University

Background: Critical visual models (e.g., identity verification, autonomous driving) are highly vulnerable to 3D adversarial attacks. Traditional defenses struggle with dynamic conditions. We propose an active defense framework that improves robustness via adaptive exploration in 3D adversarial settings.

- Implemented a **differentiable rendering environment based on EG3D** to support consistent multi-view image generation, enabling more accurate 3D representation.
- Developed an active perception defense system that integrates an **RL-based policy module** and a **perception module** in a feedback loop, enhancing environmental exploration and robustness.
- Achieved up to a **95% reduction** in attack success rate (ASR) against various adversarial attacks, surpassing passive defenses and demonstrating effectiveness in 3D adversarial settings.

Achievement: Source code is available at: https://github.com/thechentr/EAD-YOLOv5

Red Teaming Multimodal Large Language Model Security Challenge — **Tsinghua University**

Research Assistant — *Jun. 2024 - Jul. 2024*

Advisor: Associate Prof. **Hang Su** in the Department of Computer Science and Technology at Tsinghua University

Background: Visual Language Models (VLMs) are vulnerable to jailbreak attacks caused by crafted inputs. Existing attack methods lack generalization and diversity. We proposed an adversarial sample generation method that significantly improves success rates in black-box scenarios.

- Developed an **adversarial samples generation method** initiated from the visual modality to enhance attack effectiveness on VLMs.
- Integrated the **COCO dataset** with **Stable Diffusion** to improve attack stealth and diversity.
- Achieved a **26.5% increase** in attack success rate (ASR) through optimized image generation and prompt tuning.

Achievement: Our team secured **the Second Place** in the Red Teaming Multimodal Large Language Model Security Challenge hosted by **CCF**.

PROJECT EXPERIENCE

Automated C++ Code Generator for Syntax Analysis — **Jilin University**

Undergraduate Researcher — *Mar. 2023 - May. 2023*

Advisor: Associate Prof. **Huaxiao Liu** in the Department of Computer Science and Technology at Jilin University

Background: Syntax analysis is crucial in compiler construction, but building efficient parsers is complex and time-consuming. To address this, we developed an automated C++ code generator for syntax analysis that converts grammar rules into highly optimized, executable code.

- Built an **automated C++ code generator for syntax analysis**, greatly reducing manual coding effort.
- Added support for **recursive descent parsing** and **LL(1) parsing**, improving code reusability and efficiency.
- Developed an **automatic prediction set calculation module**, easing integration with various grammar rules.

Achievement: Source code is available at: https://github.com/thechentr/SNL-compiler

AWARDS, SCHOLARSHIPS & LEADERSHIP

- Outstanding Undergraduate Graduate (**Top 5%**) *2024*
- Academic Scholarship (**Top 10%**) *2021, 2022, 2023*
- **Top 8.38%** of CCF Certified Software Professional in C++ in China *2021*
- Provincial **Second Prize** in the National Undergraduate Mathematical Contest in Modeling *2022*
- Outstanding Student Leader (**Top 3%**) *2021, 2022, 2023*
- Cultural and Sports Activities Scholarship (**Top 10%**) *2022, 2024*
- Social Work Scholarship (**Top 10%**) *2023*
- **President** of the Student Union, School of Computer Science and Technology *Sept. 2022–Sept. 2023*
- **Monitor** of Class 31 of 2020, School of Computer Science and Technology *Sept. 2020–Jul. 2024*

TECHNICAL PROFICIENCIES

Programming: C/C++, Python (PyTorch), Java, R

Large Model Training: Distributed Training (DeepSpeed ZeRO-2/3), Mixed Precision, Flash Attention, PEFT, Instruction Tuning, Knowledge Distillation

ML Frameworks & Ecosystem: PyTorch, HuggingFace (Transformers, Datasets), Weights & Biases

Systems & Tools: Git, Docker, Linux, LaTeX